

ROBERT VOKÁČ, BSc

C++ Systems Engineer (Runtime / Compatibility Layers)

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PROFILE

C++ engineer focused on **low-level systems, runtime design, and API compatibility layers**.

Designed and implemented **custom runtimes replacing XNA, DirectX 3, and WinAPI**, enabling legacy applications and games to run without original dependencies.

Experience includes **reverse-engineering behavior, reproducing runtime semantics, and building cross-platform abstractions over SDL3/OpenGL**.

Transitioning from Java backend development to **full-time C++ systems engineering**, with focus on runtime and compatibility layers.

Seeking roles in **systems engineering, runtime/framework development, game technology, or backend/platform architecture**.

CORE PROJECTS

CNA

GitHub: github.com/openeggbert/cna

Custom C++ runtime reimplementing XNA 4.0 API over SDL3 (~5k LOC, production-tested on real game)

- Designed **compatibility layer matching XNA runtime behavior** (game loop, input, rendering pipeline)
- Implemented **rendering abstraction** supporting SDL_Renderer, OpenGL, and bgfx backends
- Built **resource lifecycle, input/audio systems, and framework architecture**
- Ported real-world game (**Speedy Blupi, 2013**) from C# / XNA to C++
- Reproduced original runtime behavior **without access to original engine internals**
- Achieved **smooth runtime performance** (~0.1% CPU usage, ~47 MB RAM)

Demo: <https://www.youtube.com/watch?v=TScL1L-dbHE>

Free Direct + Free API

GitHub: github.com/openeggbert/free-direct · github.com/openeggbert/free-api

Custom reimplementation of DirectX 3 (2D) and WinAPI subset in modern C++ over SDL3 (~3k LOC per module)

- Implemented **DirectDraw-style 2D rendering and WinAPI-style event/input handling**.
- Built compatibility layer enabling execution of **legacy Windows applications without original dependencies**.
- Investigating and resolving memory management issues in long-running scenarios
- Supports execution of:
 - **Speedy Blupi (1998/2001)** without DirectX SDK or WinAPI
 - **Planet Blupi (1996)** (partial compatibility, ongoing debugging)

Demo: <https://www.youtube.com/watch?v=TI3vNrpgtbo>

Hive (archived)

GitHub: github.com/robertvokac/hive

40k+ LOC modular backend platform

- Designed **metadata-driven architecture, custom ORM, and REST generation**
- Implemented **plugin system, scheduler, triggers, and validation pipeline**

SKILLS

Languages: C++20/23 (primary), SQL

Systems & Tools: SDL3, OpenGL, CMake, Linux, SQLite, Git

Core Areas:

- Runtime and framework architecture
- API compatibility layers (XNA / DirectX / WinAPI)
- Rendering and system integration
- Modular systems and abstraction design
- Debugging complex systems and legacy behavior

EXPERIENCE

Software Engineer – Assist (2017 - present)

- Maintained and improved **production backend systems (Java, SQL)**
- Diagnosed and resolved **production issues, improving system stability**
- Applied **system design principles in long-lived codebases**
- Developed **C++ systems independently (runtime and compatibility layers)**

Technologies: Java, SQL, Git, Bash

EDUCATION

BSc in Computer Science

Czech University of Life Sciences Prague (2014-2017)

LANGUAGES

English - B2

Czech - native